



Solution Guide: Network Addressing

To complete this activity, you had to:

- Convert binary traffic into IP addresses.
- Determine if they are public or private addresses.
- Compare the IPs to a list of Acme's servers to see which systems the hacker is attempting to access.

IP Address 1:

- Use the [Browserling web tool](#) to convert the numeric IP representation of 11000000101010000100010110010001 to 192.168.69.145.
- This is a private IP address and falls in the network range of Acme Trade Secrets.

IP Address 2:

- The numeric IP representation of 000010100000000000000000101010 is 10.0.0.42.
- This is a private IP address and falls in the network range of Acme Trade Secrets.

IP Address 3:

- The numeric IP representation of 11000000101011000100010110010001 is 192.172.69.145.
- This is a public IP address and falls in the network range of Acme Trade Secrets.

IP Address 4:

- The numeric IP representation of 00101001001011011011011000100000 is 41.45.182.32.
- This is a public IP address and falls in the network range of Acme Intellectual Property Secrets.

IP Address 5:

- The numeric IP representation of 0000101000000000000000001001100 is 10.0.0.76.
- This is a private IP address and falls in the network range of Acme Trade Secrets.

Final Summary

- Based on the findings, the hacker is primarily trying to access Acme Corp's trade secrets and a server containing intellectual property data.

Bonus

- The binary data of 100010001111011111000111011001010001101000110110 is actually a binary representation of a physical MAC address.
- This is likely the MAC address of the machine the hacker is trying to access or even the machine belonging to the hacker.
- MAC addresses are displayed in hex, so this needs to be converted with a binary to hex converter, such as the following: [Rapid Tables binary to hex converter](#).
- When converted, the hex representation is 88F7C7651A36 or 88:F7:C7:65:1A:36.